

Qatar-1b

R-2580

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_110610 · created 2026-08-27T00:44:18+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2455722.89499 +/- 0.00045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.169 +/- 0.012
Transit depth (Rp/Rs)^2	2.86 +/- 0.42 [%]
Orbital Inclination (inc)	82.6 +/- 0.4
Airmass coefficient 1 (a1)	1.1355 +/- 0.0089
Airmass coefficient 2 (a2)	-0.104 +/- 0.024
Scatter in the residuals of the lightcurve fit is	0.78 %
Best Comparison Star	None
Optimal Aperture	3.1
Optimal Annulus	6.41
Transit Duration (day)	0.0619 +/- 0.0033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.169 $\pm$ 0.012 vs 0.1463 (deviation 1.83 σ)
Duration fit vs published	0.0619 d vs 0.0692 d (deviation 2.13 σ)
Mid-time O–C	1.0 min
Depth SNR	13.6
Residual scatter	0.78 %

Light curve

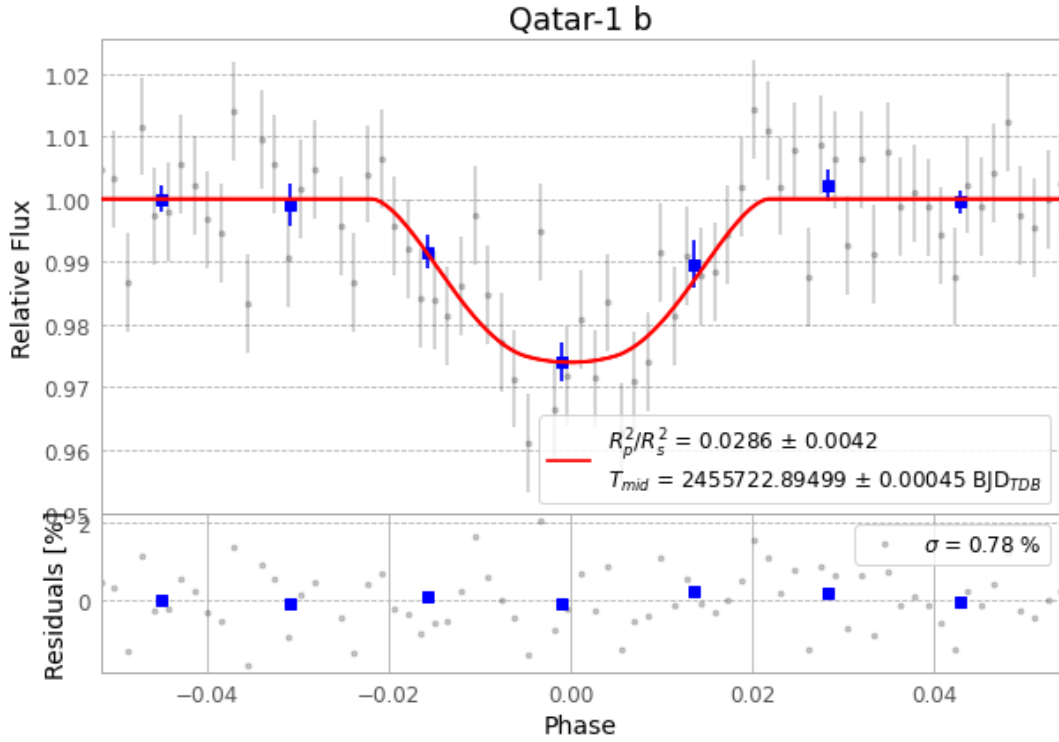


Figure 1 — FinalLightCurve_Qatar-1 b_2011-06-10.png (SHA-256 97c5ae6d5968c2ba...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [414, 255], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5d51976563f95db6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

73 FITS frames (48 MB), Qatar-1110610074212.FITS → Qatar-1110610111813.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-110610.log (462222ee2efc...), FinalLightCurve_Qatar-1 b_2011-06-10.png (97c5ae6d5968...), FinalLightCurve_Qatar-1 b_2011-06-10.csv (986684d56d4d...)

Compute resources

Wall time 118.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-27 00:44Z.